

Strategy Schema documentation v2.0

VOW Platform — Strategy Schema (Revised v2.0)

Aligned to the confirmed CTV-first agentic flow (v5)

Original version: 1.1.0 by Kareem **This revision:** 2.0.0 — reordered, scoped to CTV, and extended with client-confirmed corrections **Status:** For client verification

How to read this document. Every section is marked:

-  **UNCHANGED** — kept exactly as Kareem wrote it
-  **CHANGED** — the concept existed but is modified (original shown for comparison)
-  **NEW** — did not exist in v1.1.0, added from client feedback
-  **REMOVED** — existed in v1.1.0 but dropped for CTV scope (kept as future scope)

The document follows the **confirmed agentic flow order**, not the existing wizard order.

1. Core Principles

 **UNCHANGED** — all three kept exactly as written.

1. **Zero-Hallucination Policy:** The agent NEVER invents strategy parameters, metrics, targeting criteria, or deal IDs. It only populates values verified against the VOW database and REST APIs.
2. **Self-Filling Form Paradigm:** The agent operates as a stateful slot-filling engine backed by LangGraph. Inputs via chat or uploaded briefs are parsed into registered Pydantic slot schemas.
3. **API-Driven Tool Execution:** Every step maps to official VOW API endpoints.

2. Business Logic

2.1 Product Attribution & Selling Locations

 **UNCHANGED**

- **On Amazon (ON_AMAZON) [Endemic]:** ASINs **required**. Enables DPV, ATC, Purchase, ROAS tracking.
- **Off Amazon (NOT_SOLD_ON_AMAZON) [Non-Endemic]:** ASINs **optional** (monitors halo sales). Ad tag conversions required for site event tracking.

2.2 Attribution Window

✓ **UNCHANGED** — 14-day post-view and post-click.

2.3 Deal Types

↻ **CHANGED** — deal types unchanged, but **inventory tiers added**.

Original deal types (kept):

Type	Price	Commitment	Can pause?
Programmatic Guaranteed (PG)	Fixed CPM, guaranteed volume	Full budget owed	No
Preferred Deals	Fixed CPM	None	Yes
Private Auctions	Floor CPM, competitive	None	Yes

✚ **NEW** — **Three inventory tiers** (the primary fork in the CTV flow):

Every deal now carries an inventory tier. This classification drives most of the downstream branching — whether reach can be forecast, whether Amazon audiences apply, and whether the deal is selectable now.

Tier	Examples	Deals	Reach forecast	Audiences
Amazon owned	Prime Video	Pre-curated, selectable now	✓ Available	Amazon audiences
3P pre-curated	Netflix, Hulu, others	Pre-curated, selectable now	✗ Not available	Their own targeting (adds CPM)

3P needs curation	Disney+, others	Rate-card CPM only; VOW curates the deal after the IO is signed	✗ Not available	Their own targeting (adds CPM)
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Why this matters: a plan spanning Prime + Netflix + Disney has three portions, each with different capabilities. The agent must handle them differently — and be honest about what it can and cannot forecast.

2.4 Audience Set Profiles

 **CHANGED** — renamed "Broad" to "Wide" per client vocabulary; added fee consequence.

Profile	Was (v1.1.0)	Now
1	Narrow (High Precision)	Narrow — highly targeted, elevated intent, higher audience fee , risk of underdelivery
2	Balanced (Recommended)	Balanced — optimal blend, the usual recommendation
3	Broad (Maximum Scale)	Wide — broad demographic/interest reach, lower fee , less precision

+ NEW note: the audience fee (VCPM) stacks on top of the deal CPM. A narrow audience is both smaller **and** more expensive per impression. The agent should surface the **effective CPM** (deal + audience fee), not just the deal price.

+ NEW: audiences are **mandatory** and **suggestion-driven**. The agent always suggests three options using VOW's existing pgvector + OpenAI feature (`P0ST /audience-sets/suggest/`). Nobody browses the ~3,400 segments manually.

✗ REMOVED for CTV: product audiences (not applicable per client). AMC audiences are **conditional** — available only when the advertiser has prior campaign data (retargeting tactic).

3. The Agentic Flow — Step by Step

 **CHANGED — entirely reordered.** The original followed the 6-step UI wizard. This follows the client-confirmed CTV-first agentic flow (v5).

Comparison: old order vs new order

Old (v1.1.0 wizard)	New (v2.0 agentic, confirmed)
1. Strategy details	1. Basics (+ durations)
2. Goal, KPI & bid	<i>(goal/KPI/bid folded into Basics)</i>
3. Deals	2. CTV inventory (three-tier fork)
—	3. Budget split + NEW
4. Audiences	4. Audiences (mandatory, suggestion-driven)
—	5. Targeting + NEW
<i>(forecast was a sub-step)</i>	6. Predict reach (Amazon only ; repair loop)
—	7. Plan approval + NEW
<i>(create was at the end)</i>	8. Create the real strategy
5. Creatives	9. Upload video creative (+ duration check)
—	10. Platform creative approval + NEW
<i>(ASINs were in step 1)</i>	11. Tracking setup (ASINs + ad tag)  MOVED
—	12. Credit check + NEW
6. Summary → create	13. Activate + NEW

Step 1: Basics

 **CHANGED** — merged original Steps 1 and 2 (strategy details + goal/KPI/bid), added durations, scoped to CTV.

What was in v1.1.0 (Step 1 + Step 2):

- Strategy name, flight dates, target markets, primary currency, formats (all four), product categories, selling location, ASINs
- Goal (three choices), KPI (six choices), ad tag conversions, market budgets, base bids

What it is now:

Field	Type	Requirement	Change from v1.1.0
Strategy name	String	Required	 Unchanged. Validated via <code>GET /api/strategies/check_strategy_name_uniqueness/</code>
Flight dates	Date range	Required	 Unchanged. <code>lower</code> ≥ today, <code>upper</code> > <code>lower</code>
Target markets	Multi-select	Required	 Unchanged. ISO country codes (GB, US, DE)
Primary currency	Dropdown	Required	 Unchanged. EUR, GBP, USD
Creative durations	Multi-select	Required	 NEW. Values: <code>10</code> , <code>15</code> , <code>20</code> , <code>30</code> (seconds). Determines which deals are available and

			what CPM applies
Goal	Fixed	Required	 CHANGED. For CTV, always Awareness. Client: "CTV is typically used as an Awareness goal as it's hard to track anything further down the funnel"
KPI	Select	Required	 CHANGED. For CTV, reach or frequency only. Was six choices; others scoped out
Market budgets	Table	Required	 Unchanged. Per-market budget, must be > 0
Base bids	Table	Required	 Unchanged. Per-market base CPM bid
Frequency cap	Number	Optional	 NEW. Was absent; client confirmed optional
Budget cap	Number	Optional	 NEW. Was absent; client confirmed optional

Formats	Fixed	Required	 CHANGED. For M1, streaming_tv and prime_video only. Display and online_video removed from scope
Product categories	Multi-select	Required for video	 Unchanged. Fetched via <code>GET /api/contextual-targeting/{market}/product-categories/</code>
Selling location	Radio	Required	 Unchanged. <code>ON_AMAZON</code> or <code>NOT_SOLD_ON_AMAZON</code>
Product ASINs	Textarea	Conditional	 MOVED. Still required if <code>ON_AMAZON</code> , but the validation and collection now happens at Step 11 (tracking setup). See open question below

API calls at this step: `GET /api/strategies/check_strategy_name_uniqueness/`,
`GET /api/contextual-targeting/{market}/product-categories/`

✗ REMOVED from this step: ad tag conversions (moved to Step 11), the three non-CTV format options (Display, Online Video — future scope), the four non-awareness KPIs (CTR, CPC, CPA,

CPDPV — future scope)

 **Open question:** `product_location` and `asin_numbers` are fields in the `POST /strategies/` payload called at Step 8. If ASINs are collected at Step 11 (after Step 8), they'd need to be patched onto the strategy afterwards. Alternatively, the ASIN question stays early (it's a plan field) and only the ad-tag check moves late. **Confirm with client.**

Step 2: CTV Inventory (the tier fork)

 **CHANGED** — was Step 3 "Deals" in v1.1.0. Now comes **before audiences**, and introduces the three-tier fork.

What was in v1.1.0:

- A flat deals table filtered by market and format, with checkbox selection

What it is now:

Field	Type	Requirement	Change from v1.1.0
Selected deals	Checkbox table	Required	 Core concept unchanged. Fetched via <code>GET /api/deals/?markets={market}&formats=streaming_tv</code>
Inventory tier (per deal)	Enum	Derived	 NEW. Each deal classified as <code>AMAZON_OWNED</code> , <code>THIRD_PARTY_PRECURATED</code> , or <code>THIRD_PARTY_NEEDS_CURATION</code>

CTV rate card	Reference	Read	+ NEW. GET /api/rates/c tv/{market}/ — channels, durations, CPMs
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+ NEW — Genre upsell logic: The client asked: "based on the brief we can suggest whether a specific available genre would be a better match at a slightly higher CPM." Example: Prime Video ROS at \$18.22 vs Action at \$22.07 — the agent should recommend when the brief implies a genre match.

+ NEW — Curation capture (for 3P-needs-curation tier): When deals can't be selected yet (Disney+ etc.), the agent captures what VOW needs to curate later: genres, durations, targeting preferences, budget, flight dates.

Field	Type	Requirement
Curation: genres	Multi-select	Required for curation tier
Curation: durations	Multi-select	Required for curation tier
Curation: targeting prefs	Text	Optional
Curation: budget	Number	Required for curation tier
Curation: flight dates	Date range	Required for curation tier

API calls at this step: GET /api/deals/, GET /api/deals/filter-properties/, GET /api/rates/ctv/{market}/

Step 3: Budget Split

+ ENTIRELY NEW — did not exist in v1.1.0. Added per client requirement.

"We will need to support the suggested budget split across inventories or creative durations."

The agent proposes how the total budget is divided across inventories (Prime / Netflix / Disney) and across creative durations (15s / 30s). This is genuinely hard — different durations have different CPMs, and there's no reach data for Netflix/Disney to optimise against.

Field	Type	Requirement
Split by inventory	Allocation (%)	Required when multiple inventories selected
Split by duration	Allocation (%)	Required when multiple durations selected
Split method	Enum	Agent states its assumption

Split method options:

- `EVEN_BY_BUDGET` — same £ per inventory/duration; **uneven impressions** (higher CPM = fewer impressions)
- `EVEN_BY_IMPRESSIONS` — same impression count; **uneven £** (higher CPM = more spend)

The agent must **state which it chose and why**, so the trader can adjust. Example: "I've split evenly by impressions, which weights spend toward the 30s at its higher CPM."

No API call — this is agent-side logic. The resulting budgets feed into the `market_budgets` field at strategy creation.

Step 4: Audiences

 **CHANGED** — was Step 4 in v1.1.0 and optional. Now mandatory, suggestion-driven, and positioned after the budget split.

What was in v1.1.0:

- Browse/search audience sets, checkbox selection, Similar/Exact toggle

What it is now:

Field	Type	Requirement	Change from v1.1.0
Audience options	3 profiles	Required	 CHANGED from optional to mandatory.

			Agent always generates narrow / balanced / wide
Chosen option	Select one	Required	+ NEW. Trader picks one of the three
Matching mode	Toggle	Required	☒ Unchanged. Similar vs Exact
Effective CPM (per option)	Display	Read-only	+ NEW. Deal CPM + audience VCPM fee, shown per option so the trader sees the real cost

Constraints for CTV:

- Amazon audiences **only apply to Amazon-owned inventory**. For Netflix/Disney, their own targeting applies
- **✗** Product audiences **not applicable** to CTV (removed)
- AMC audiences are **conditional** — only when the advertiser has prior campaign data
- **Nobody browses** — the agent uses `POST /api/audience-sets/suggest/` exclusively
- The audience set does **not** need to be created before forecasting — it's created later at strategy creation via a simplified CTV endpoint

API calls at this step: `POST /api/audience-sets/suggest/` → `GET /api/audience-sets/suggest/{id}/`

⚠ Open question: the suggest endpoint's response shape. v1.1.0 assumed it returns `bundles.narrow/balanced/broad`. The real endpoint may return a flat list that we group ourselves. **Confirm against the real API.**

Step 5: Targeting

+ ENTIRELY NEW — did not exist in v1.1.0.

Field	Type	Requirement
Location	Multi-select	Optional
Instream position	Select	Optional
Content-category exclusions	Multi-select	Optional
Device type	Multi-select	Optional
Mobile environment	Select	Optional

Critical design note from the client: *"This targeting list frequently changes so it should be easy to add new targeting types."* — the implementation must be **config-driven**, not hard-coded. Adding a new targeting type should be a configuration change, not a code change.

Not supported by VOW today (future scope): genre exclusions, day-parting, language.

API calls at this step: `POST /api/contextual-targeting/{market}/products/` , `GET /api/strategies/locations/{market}/`

Step 6: Predict Reach

 **CHANGED** — was embedded in the original flow. Now a first-class step with the tier-based honesty rule.

Field	Type	Requirement	Change from v1.1.0
Reach curve	Chart	Read-only (Amazon only)	 CHANGED. Only available for Amazon-owned inventory. For 3P, state honestly that reach is unavailable
Estimated impressions	Number	Read-only	 Unchanged
Estimated unique reach	Number	Read-only (Amazon only)	 CHANGED. Not available for

			Netflix/Disney
Average frequency	Number	Read-only (Amazon only)	 CHANGED. Not available for Netflix/Disney
Indicative CPM	Number	Read-only	 Unchanged

+ NEW — the honesty rule for 3P inventory: For Netflix/Disney, the agent shows: rate-card CPM and **derived impressions** ($\text{budget} \div \text{CPM} \times 1,000$). It explicitly states that reach is unavailable and why. **Never invent a reach number.**

+ NEW — consequences:

- The **repair loop** (too narrow → widen → re-forecast) applies **only to the Amazon portion**
- **Total reach cannot be summed** across providers (no cross-platform deduplication)

Repair loop (v1.1.0 §7.1 —  concept correct,  mechanism updated):

Was (v1.1.0)	Now
If <code>estimated_unique_reach == 0</code> , switch from Narrow to Balanced/Broad	If reach is insufficient, extend the audience (not necessarily switch profiles — could add segments within the chosen profile)
Also adjust base CPM bid upward	 Still valid as Action 2
Re-run forecast	 Unchanged

API calls at this step: `POST /api/audience-sets/reach-forecast/` (or the simplified CTV endpoint, name TBC)

Step 7: Plan Approval

+ ENTIRELY NEW — did not exist in v1.1.0.

*The client confirmed: approval gates the **plan**, before it is finalised. Not before launch. Optionally routes to a manager.*

Field	Type	Requirement
Approval status	Enum	Required

Approved by	String (user)	Set on approval
Approved at	Timestamp	Set on approval
Manager required	Boolean	Configurable (possibly budget-threshold-based)
Rejection reason	Text	Required on reject

Values: PENDING → APPROVED or REJECTED

Implementation: LangGraph `interrupt()`. The graph physically stops and persists state. It cannot proceed until a human sends approve or reject. The budget is locked at this moment — nothing launches that a person hasn't approved.

On rejection: the flow returns to Step 4 (audiences) so the trader can adjust the plan.

No API call — this is agent-internal. The approval is logged in the audit trail.

Step 8: Create the Real Strategy

 **CHANGED** — was "Summary & Create" (Step 6) in v1.1.0. Key change: create the **real** strategy, not a draft.

What was in v1.1.0:

- Summary view → `POST /api/strategies/` or `POST /api/strategies/draft/` → returns `status: "draft"`

What it is now:

Field	Change
Endpoint	 <code>POST /api/strategies/</code> — not <code>/strategies/draft/</code> . Client: "don't need to create draft strategy; draft is just for the wizard creation"
Audience set	 Created at this step via the simplified CTV endpoint (not before forecasting)

All slots	All filled slots from Steps 1–7 are assembled into the creation payload
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API calls at this step: `POST /api/strategies/` , audience-set creation via CTV endpoint

⚠ **Open question:** what status does the created strategy land in? If it's still "draft" by default, activation via `set_status` remains a separate step. **Confirm with client.**

Step 9: Upload Video Creative

🔄 **CHANGED** — was Step 5 "Creatives" in v1.1.0. Simplified to video only, moved to after plan approval, and duration check added.

What was in v1.1.0:

- Browse assets and pre-approved creatives, select from table, add click-through URL

What it is now:

Field	Type	Requirement	Change from v1.1.0
Video file	Upload (direct or URL)	Required	🔄 CHANGED. For CTV, always video . No display creatives, no pre-approved selection, no responsive e-commerce
Click-through URL	URL	Required	✅ Unchanged
Duration	Derived from file	Checked	➕ NEW. Must match one of the durations in the approved plan

➕ **NEW — Duration match check:** If the uploaded video is 30s but the approved plan specified 15s deals, the economics change (different CPM → different impressions for the same budget).

This triggers **re-approval** (return to Step 7 with the amended plan).

Upload path: POST /api/assets/amz_assets/gen_upload_urls/ (get upload URLs) →
POST /api/assets/amz_assets/register/ (register the asset on Amazon)

✗ **REMOVED for CTV:** browse existing assets (GET /api/assets/), pre-approved creatives (GET /api/creatives/), responsive e-commerce (POST /api/creatives/recs/), third-party tags (POST /api/creatives/third-party/). These are valid for Display but not for CTV scope.

Step 10: Platform Creative Approval

✚ **ENTIRELY NEW** — did not exist in v1.1.0.

Field	Type	Requirement
Amazon approval status	Enum	Read-only
Netflix approval status	Enum	Read-only (if Netflix inventory)
Disney approval status	Enum	Read-only (if Disney inventory)

Values: PENDING → APPROVED or REJECTED

Every video must pass the platform's content and technical review before it can run. Each platform reviews its own inventory independently. A plan can be fully approved and funded and still not launch until the creative clears.

On rejection: the agent reports the reason and asks for a replacement (return to Step 9).

⚠ **Open question:** do Netflix/Disney review statuses surface inside VOW's API, or is that tracked externally? **Confirm with client.**

Step 11: Tracking Setup

↻ **MOVED** — ASIN validation was in Step 1 (strategy details) and ad-tag conversions were in Step 2 (goal/KPI). Both now sit here, after creative approval and before tracking is attached.

What was in v1.1.0:

- ASINs collected in Step 1 and validated via `POST /api/contextual-targeting/{market}/asin-validation/`
- Ad tag conversions selected in Step 2 via `GET /api/conversions/definitions/`

What it is now:

Field	Type	Requirement	Change from v1.1.0
Sells on Amazon?	Question	Asked here	 MOVED from Step 1
Product ASINs	Textarea	Required if endemic	 Validation unchanged: <code>POST /api/contextual-targeting/{market}/asin-validation/</code>
Sells on own website?	Question	Asked here	 NEW explicit question
Ad tag registered?	Check	Required if yes	 NEW. Check whether an ad tag is already registered. If not, show setup instructions — the tag must be installed before the campaign runs (tracking only records activity after it goes live)
Ad tag conversions	Multi-select	Required if ad tag exists	 MOVED from Step 2. Events: Page view, Add

			to cart, Checkout, Application. Via GET /api/convers ions/definit ions/
--	--	--	---

API calls at this step: POST /api/contextual-targeting/{market}/asin-validation/, GET /api/conversions/definitions/

⚠ **Open question (repeated from Step 1):** since product_location and asin_numbers are fields in POST /strategies/ (called at Step 8), they may need to be collected before Step 8 and only the ad-tag check moves here. **Confirm with client.**

Step 12: Credit Check

✚ **ENTIRELY NEW** — did not exist in v1.1.0.

Credit is checked **only at activation**, not during planning. Everything before this point is a costless plan.

Field	Type	Requirement
Account balance	Number	Read-only
Strategy budget	Number	Read-only
Sufficient	Boolean	Derived (balance ≥ budget)

If insufficient: prompt a top-up via POST /api/credits/ or POST /api/credits/stripe/.

API call: GET /api/credits/summary/

Step 13: Activate

✚ **ENTIRELY NEW** — did not exist in v1.1.0 (was implicit in "create strategy").

The single spend action in the entire flow. Everything before this was free.

API call: POST /api/strategies/{id}/set_status/

After activation, VOW's outbound sync creates the Campaigns and Ad Groups on Amazon DSP.

4. API Catalogue

 **CHANGED** — original catalogue kept, with additions and removals marked.

Operation	Method	Endpoint	Status
Check name uniqueness	GET	/api/strategies/check_strategy_name_uniqueness/	✓ Unchanged
ASIN validation	POST	/api/contextual-targeting/{market}/asin-validation/	✓ Unchanged
Product categories	GET	/api/contextual-targeting/{market}/product-categories/	✓ Unchanged
Conversion definitions	GET	/api/conversions/definitions/	✓ Unchanged
List deals	GET	/api/deals/	✓ Unchanged
Deal filter properties	GET	/api/deals/filter-properties/	✓ Unchanged
List audience sets	GET	/api/audience-sets/	✓ Unchanged
Suggest audiences	POST	/api/audience-	✓ Unchanged

		sets/suggest/ 	
Audience reach forecast	POST	/api/audience-sets/reach-forecast/	✓ Unchanged
Strategy reach forecast	POST	/api/strategies/reach-forecast/	✓ Unchanged
List assets	GET	/api/assets/	✓ Unchanged
List creatives	GET	/api/creatives/	✓ Unchanged
Create strategy	POST	/api/strategies/	✓ Unchanged
Read strategy	GET	/api/strategies/{id}/	✓ Unchanged
CTV rate card	GET	/api/rates/ctv/{market}/	+ NEW
Inventory sources	GET	/api/inventory-sources/	+ NEW
Activate strategy	POST	/api/strategies/{id}/set_status/	+ NEW
Credit summary	GET	/api/credits/summary/	+ NEW
Upload URLs	POST	/api/assets/amz_assets/generate_upload_urls/	+ NEW

Register asset	POST	/api/assets/ amz_assets/r egister/	+ NEW
Locations	GET	/api/strateg ies/location s/{market}/	+ NEW
Draft create	POST	/api/strateg ies/draft/	✗ REMOVED — client: "draft is just for the wizard"

5. Pydantic Data Models

 **CHANGED** — original models kept where valid, extended and restructured.

```

1 from enum import Enum
2 from typing import Optional
3 from pydantic import BaseModel, Field, HttpUrl
4
5
6 # =====
7 # ENUMS
8 # =====
9
10 class ChannelTypeEnum(str, Enum):
11     """ UNCHANGED"""
12     DSP = "dsp"
13     SPONSORED = "sponsored"
14
15 class GoalEnum(str, Enum):
16     """ CHANGED — kept all values, but for CTV M1 only AWARENESS is
17     used"""
18     AWARENESS = "AWARENESS"
19     CONSIDERATION = "CONSIDERATION" # future scope
20     CONVERSION = "CONVERSION" # future scope
21
22 class KPIEnum(str, Enum):
23     """ CHANGED — kept all values, but for CTV M1 only reach and
24     frequency"""
25     REACH = "reach"
26     FREQUENCY = "frequency"
27     CTR = "ctr" # future scope
28     CPC = "cpc" # future scope
29     CPA = "cpa" # future scope
30     CPDPV = "cpdpv" # future scope
31
32 class ProductLocationEnum(str, Enum):
33     """ UNCHANGED"""
34     ON_AMAZON = "ON_AMAZON"
35     NOT_SOLD_ON_AMAZON = "NOT_SOLD_ON_AMAZON"
36
37 class FormatEnum(str, Enum):

```

```

36     """🔄 CHANGED – kept all values, but for CTV M1 only streaming_tv
and prime_video"""
37     DISPLAY = "display" # future scope
38     ONLINE_VIDEO = "online_video" # future scope
39     STREAMING_TV = "streaming_tv"
40     PRIME_VIDEO = "prime_video"
41
42     class CurrencyEnum(str, Enum):
43         """✅ UNCHANGED"""
44         EUR = "EUR"
45         GBP = "GBP"
46         USD = "USD"
47
48     # + NEW ENUMS
49
50     class DurationEnum(str, Enum):
51         """+ NEW – creative durations for CTV"""
52         TEN = "10"
53         FIFTEEN = "15"
54         TWENTY = "20"
55         THIRTY = "30"
56
57     class InventoryTierEnum(str, Enum):
58         """+ NEW – the three inventory tiers driving the flow's primary
fork"""
59         AMAZON_OWNED = "AMAZON_OWNED"
60         THIRD_PARTY_PRECURATED = "THIRD_PARTY_PRECURATED"
61         THIRD_PARTY_NEEDS_CURATION = "THIRD_PARTY_NEEDS_CURATION"
62
63     class ApprovalStatusEnum(str, Enum):
64         """+ NEW – for plan approval and creative approval"""
65         PENDING = "PENDING"
66         APPROVED = "APPROVED"
67         REJECTED = "REJECTED"
68
69     class BudgetSplitMethodEnum(str, Enum):
70         """+ NEW – how the budget is divided"""
71         EVEN_BY_BUDGET = "EVEN_BY_BUDGET"
72         EVEN_BY_IMPRESSIONS = "EVEN_BY_IMPRESSIONS"
73         CUSTOM = "CUSTOM"
74
75     class AudienceProfileEnum(str, Enum):
76         """+ NEW – the three audience options"""
77         NARROW = "NARROW"
78         BALANCED = "BALANCED"
79         WIDE = "WIDE"
80
81
82     # =====
83     # COMPONENT SCHEMAS
84     # =====
85
86     class DateRangeSchema(BaseModel):
87         """✅ UNCHANGED"""
88         lower: str = Field(..., description="ISO date YYYY-MM-DD")
89         upper: str = Field(..., description="ISO date YYYY-MM-DD")
90         bounds: str = Field(..., description="Interval boundary
notation")
91
92     class MarketBudgetBidSchema(BaseModel):
93         """✅ UNCHANGED"""
94         market: str = Field(..., description="ISO country code")
95         budget: str = Field(..., description="Total budget decimal
string")
96         base_bid: str = Field(..., description="Base CPM bid decimal
string")
97

```

```

98 class SelectedDealSchema(BaseModel):
99     """🔄 CHANGED – added inventory_tier, genre, ad_lengths,
    provider"""
100     deal_id: str = Field(..., description="External deal ID e.g.
    EXT7P75718S8MNR")
101     name: str = Field(..., description="Deal name")
102     cpm: str = Field(..., description="Fixed or floor CPM price")
103     inventory_tier: InventoryTierEnum = Field(..., description="Which
    tier this deal belongs to") # + NEW
104     provider: str = Field(..., description="e.g. Prime Video,
    Netflix, Disney+") # + NEW
105     genre: Optional[str] = Field(None, description="Genre if genre-
    specific deal") # + NEW
106     ad_lengths: list[str] = Field(default_factory=list,
    description="Supported durations") # + NEW
107     deal_type: str = Field(..., description="PG, Preferred, or
    Private Auction") # + NEW
108
109 class SelectedAudienceSetSchema(BaseModel):
110     """🔄 CHANGED – added profile and effective_cpm"""
111     audience_set_id: str = Field(..., description="Audience set
    UUID")
112     name: str = Field(..., description="Audience set name")
113     vcpm_fee: str = Field(..., description="VCPM fee decimal")
114     profile: AudienceProfileEnum = Field(..., description="Narrow,
    Balanced, or Wide") # + NEW
115     effective_cpm: Optional[str] = Field(None, description="Deal CPM
    + audience VCPM") # + NEW
116     estimated_reach: Optional[int] = Field(None, description="If
    Amazon inventory") # + NEW
117
118 class SelectedCreativeSchema(BaseModel):
119     """🔄 CHANGED – added duration_seconds for the match check"""
120     asset_id: str = Field(..., description="Registered asset ID")
121     click_through_url: HttpUrl = Field(..., description="Landing page
    URL")
122     duration_seconds: int = Field(..., description="Video length in
    seconds") # + NEW
123     upload_method: str = Field("direct", description="direct or url")
    # + NEW
124
125 # + NEW SCHEMAS
126
127 class BudgetSplitSchema(BaseModel):
128     """+ NEW – how budget is divided across inventories and
    durations"""
129     method: BudgetSplitMethodEnum = Field(..., description="Even by
    budget, even by impressions, or custom")
130     by_inventory: list[dict] = Field(..., description="[{provider,
    budget, impressions_estimate}]")
131     by_duration: list[dict] = Field(..., description="[{duration,
    budget, cpm, impressions_estimate}]")
132
133 class CurationRequirementsSchema(BaseModel):
134     """+ NEW – captured for 3P-needs-curation inventory (e.g.
    Disney+)"""
135     provider: str = Field(..., description="e.g. Disney+")
136     genres: list[str] = Field(default_factory=list)
137     durations: list[str] = Field(default_factory=list)
138     targeting_preferences: Optional[str] = None
139     budget: str = Field(..., description="Allocated budget for this
    provider")
140     flight_dates: DateRangeSchema = Field(...)
141
142 class TargetingSchema(BaseModel):
143     """+ NEW – CTV targeting options (config-driven, extensible)"""
144     locations: list[str] = Field(default_factory=list)

```

```

145     instream_positions: list[str] = Field(default_factory=list)
146     content_category_exclusions: list[str] =
Field(default_factory=list)
147     device_types: list[str] = Field(default_factory=list)
148     mobile_environments: list[str] = Field(default_factory=list)
149
150     class ForecastResultSchema(BaseModel):
151         """🔄 CHANGED – added availability flag for the honesty rule"""
152         is_available: bool = Field(..., description="False for
Netflix/Disney – no reach data") # + NEW
153         estimated_impressions: Optional[int] = None
154         estimated_unique_reach: Optional[int] = Field(None,
description="Only for Amazon inventory")
155         average_frequency: Optional[float] = Field(None,
description="Only for Amazon inventory")
156         indicative_cpm: Optional[str] = None
157         reach_curve: Optional[list[dict]] = Field(None, description="
[{budget, reach}] – Amazon only")
158
159     class TrackingSetupSchema(BaseModel):
160         """+ NEW – tracking prerequisites collected at Step 11"""
161         sells_on_amazon: bool = Field(...)
162         validated_asins: list[dict] = Field(default_factory=list,
description="[{'asin', 'title', 'brand'}]")
163         sells_on_own_site: bool = Field(...)
164         ad_tag_registered: Optional[bool] = None
165         ad_tag_conversions: list[str] = Field(default_factory=list,
description="Selected conversion events")
166
167
168     # =====
169     # FULL STRATEGY SCHEMA
170     # =====
171
172     class FullStrategySchema(BaseModel):
173         """🔄 CHANGED – restructured from wizard steps to semantic
grouping"""
174
175         # --- Identity ---
176         id: Optional[str] = Field(None, description="System-assigned
strategy ID")
177         advertiser_id: str = Field(..., description="Parent advertiser
UUID")
178         channel_type: ChannelTypeEnum = ChannelTypeEnum.DSP
179
180         # --- Basics (Step 1) ---
181         name: str = Field(..., description="Unique strategy name")
182         flight_dates: DateRangeSchema = Field(...)
183         markets: list[str] = Field(..., description="ISO country codes")
184         primary_currency: CurrencyEnum = Field(CurrencyEnum.GBP)
185         durations: list[DurationEnum] = Field(..., description="Creative
durations") # + NEW
186         formats: list[FormatEnum] = Field(...)
187         goal: GoalEnum = Field(GoalEnum.AWARENESS, description="Fixed for
CTV") # 🔄 CHANGED default
188         kpi_target_type: KPIEnum = Field(...)
189         product_categories: list[int] = Field(default_factory=list)
190         product_location: ProductLocationEnum = Field(...)
191         market_budgets: list[MarketBudgetBidSchema] = Field(...)
192         frequency_cap: Optional[int] = Field(None, description="Optional
weekly cap") # + NEW
193         budget_cap: Optional[str] = Field(None, description="Optional
budget cap") # + NEW
194
195         # --- Inventory (Step 2) ---
196         selected_deals: list[SelectedDealSchema] = Field(...) # 🔄
CHANGED – enriched schema

```



```

197     curation_requirements: list[CurationRequirementsSchema] =
Field(default_factory=list) # + NEW
198
199     # --- Budget Split (Step 3) ---
200     budget_split: Optional[BudgetSplitSchema] = None # + NEW
201
202     # --- Audiences (Step 4) ---
203     audience_options: list[SelectedAudienceSetSchema] =
Field(default_factory=list) # CHANGED - now carries all three
204     chosen_audience_profile: Optional[AudienceProfileEnum] = None #
+ NEW
205     matching_mode: str = Field("Exact", description="Similar or
Exact") # UNCHANGED
206
207     # --- Targeting (Step 5) ---
208     targeting: Optional[TargetingSchema] = None # + NEW
209
210     # --- Forecast (Step 6) ---
211     forecast: Optional[ForecastResultSchema] = None # CHANGED -
enriched with availability
212
213     # --- Approval (Step 7) ---
214     approval_status: Optional[ApprovalStatusEnum] = None # + NEW
215     approved_by: Optional[str] = None # + NEW
216     approved_at: Optional[str] = None # + NEW
217
218     # --- Creative (Step 9) ---
219     selected_creatives: list[SelectedCreativeSchema] =
Field(default_factory=list) # CHANGED - enriched
220     creative_duration_match: Optional[bool] = None # + NEW
221     creative_approval_status: Optional[ApprovalStatusEnum] = None #
+ NEW
222
223     # --- Tracking (Step 11) ---
224     tracking: Optional[TrackingSetupSchema] = None # + NEW
225     product_asins: list[str] = Field(default_factory=list) #
MOVED from Step 1
226
227     # --- Activation (Steps 12-13) ---
228     credit_sufficient: Optional[bool] = None # + NEW
229     status: str = Field("created", description="Strategy status") #
CHANGED from "draft"
230     is_syncing: bool = Field(False)
231
232
233     # =====
234     # LANGGRAPH PLANNING STATE
235     # =====
236
237     # CHANGED - restructured from wizard-step-based to semantic field
names
238
239     # WAS (v1.1.0):
240     # class PlanningAgentState(TypedDict):
241     #     messages: List[Dict[str, Any]]
242     #     advertiser_id: str
243     #     current_step: int # 0 to 5
244     #     strategy_id: Optional[str]
245     #     step1_details: Optional[Dict[str, Any]]
246     #     step2_goal_kpi_bid: Optional[Dict[str, Any]]
247     #     step3_deals: Optional[Dict[str, Any]]
248     #     step4_audiences: Optional[Dict[str, Any]]
249     #     step5_creatives: Optional[Dict[str, Any]]
250     #     forecast_results: Optional[Dict[str, Any]]
251     #     validation_errors: List[str]
252     #     is_complete: bool
253

```

```

254 # NOW:
255 from typing import Annotated
256 from typing_extensions import TypedDict
257 from langgraph.graph.message import add_messages
258
259 class PlanningAgentState(TypedDict):
260     """State carried through the LangGraph planning flow.
261
262     Named semantically, not by wizard step – the state describes
263     the plan, not the UI that collected it.
264     """
265     # --- Conversation ---
266     messages: Annotated[list, add_messages]
267
268     # --- Session context ---
269     advertiser_id: str
270     session_id: str
271     current_stage: str # + NEW – for
272     the adaptive canvas
273     current_artifact_id: Optional[str] # + NEW – for
274     the adaptive canvas
275
276     # --- Basics ---
277     strategy_name: Optional[str]
278     flight_dates: Optional[dict]
279     markets: list[str]
280     durations: list[str] # + NEW
281     primary_currency: str
282     goal: str # fixed:
283
284     AWARENESS for CTV
285     kpi: str # reach or
286     frequency
287     market_budgets: list[dict]
288     product_location: Optional[str]
289     frequency_cap: Optional[int] # + NEW
290     budget_cap: Optional[str] # + NEW
291
292     # --- Inventory ---
293     inventory_tier: Optional[str] # + NEW – which
294     tier fork we're on
295     selected_deals: list[dict]
296     curation_requirements: list[dict] # + NEW
297
298     # --- Budget split ---
299     budget_split: Optional[dict] # + NEW
300
301     # --- Audiences ---
302     audience_options: list[dict] # the three
303     profiles
304     chosen_audience: Optional[dict] # which one the
305     trader picked
306
307     # --- Targeting ---
308     targeting: Optional[dict] # + NEW
309
310     # --- Forecast ---
311     forecast: Optional[dict] #
312     reach/impressions/CPM (with availability flag)
313
314     # --- Approval ---
315     approval_status: Optional[str] # + NEW –
316     PENDING/APPROVED/REJECTED
317     approved_by: Optional[str] # + NEW
318     approved_at: Optional[str] # + NEW
319
320     # --- Creative ---
321     creative_id: Optional[str]

```

```

312 creative_duration_match: Optional[bool] # + NEW
313 creative_approval_status: Optional[str] # + NEW
314
315 # --- Tracking ---
316 tracking_setup: Optional[dict] # + NEW
317 product_asins: list[str] # ↻ MOVED
318
319 # --- Activation ---
320 credit_sufficient: Optional[bool] # + NEW
321 strategy_id: Optional[str]
322 strategy_status: Optional[str]
323
324 # --- Errors ---
325 validation_errors: list[str]
326

```

6. State Machine

 **CHANGED — needs complete rebuild.** The original was a linear pipe. The confirmed flow has branches, loops, and interrupts.

The confirmed state machine (v5):

```

1  START
2    → extract_fields (slot-filling from brief)
3    → select_inventory (CTV, three-tier fork)
4      → [if 3P needs curation] capture_curation_requirements
5    → propose_budget_split (across inventories + durations)
6    → suggest_audiences (3 options via pgvector; mandatory)
7    → apply_targeting (optional, configurable)
8    → predict_reach
9      → [if Amazon] real forecast + reach curve
10     → [if 3P] CPM + derived impressions only (honest)
11     → [if too narrow] REPAIR: extend audience → re-predict (loop)
12    → present_plan (on the strategy card)
13    →  PLAN APPROVAL (interrupt – optionally a manager)
14      → [if rejected] return to suggest_audiences
15    → create_strategy (POST /strategies/ – the real one, not draft)
16    → upload_creative (video, gen_upload_urls + register)
17      → [if duration mismatch] amend plan → RE-APPROVE (loop back)
18    → platform_creative_approval (Amazon / Netflix / Disney review)
19      → [if rejected] return to upload_creative
20    → tracking_setup (ASINs + ad tag check)
21    → credit_check (GET /credits/summary/)
22      → [if insufficient] prompt top-up (loop)
23    → activate (POST /strategies/{id}/set_status/ – the single spend
24      action)
25    → DONE

```

Q&A side path: at any point, the trader can ask a pricing/availability question ("what's the CPM for Netflix 30s?"). The agent answers from the rate card and resumes.

7. Brief Parsing & Edge Cases

7.1 Entity Normalisation

 **UNCHANGED** — the original examples are correct. **Additions:**

Input	Extraction	Status
August 2026	flight_dates: {lower: "2026-08-01", upper: "2026-08-31"}	✓ Original
UK	markets: ["GB"], primary_currency: "GBP"	✓ Original
£10,000	market_budgets: [{market: "GB", budget: "10000.00"}]	✓ Original
education website	product_location: "NOT_SOLD_ON_AMAZON"	✓ Original
30 seconds	durations: ["30"]	+ NEW
UK and France	markets: ["GB", "FR"]	+ NEW
sports drink	Consider genre-specific deals (Sports)	+ NEW
Prime and Netflix	Multiple inventory tiers	+ NEW

7.2 Validation Failure Protocols

✓ **UNCHANGED** — duplicate name, invalid ASIN, past dates protocols all correct.

7.3 Repair Loop

↻ **CHANGED** — concept correct, mechanism updated (see Step 6 above). Only applies to Amazon-owned inventory.

+ **NEW** — **"Did I understand correctly?" confirmation.** After extracting fields from a brief, the agent immediately shows what it understood so the trader can correct before proceeding. This

is the single most important trust mechanism in the product.

8. Summary of all changes

Category	Count	Items
✓ Unchanged	~15	Core principles, product attribution, deal types, date validation, name uniqueness, currency, most API endpoints, brief parsing examples
↻ Changed	~12	Step order, goal scoped to Awareness, KPI scoped to reach/frequency, deals enriched with tier, audiences mandatory + renamed Wide, forecast with availability flag, state restructured, creative simplified to video
+ New	~15	Durations, inventory tiers, budget split, targeting, plan approval, creative duration check, platform creative approval, tracking setup (moved), credit check, activation, curation capture, effective CPM, adaptive-canvas fields

 Removed	~5	Draft endpoint, product audiences, non-CTV formats (scoped out), non-awareness KPIs (scoped out), canary-check
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This document is for client verification. Once confirmed, it becomes the shared contract that Wajahat (state + graph), Vishal (registry), and Basil (adaptive canvas) build against.